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Redshift-Agnostic Machine Learning Classification: Unveiling Peak Performance in Galaxy, Star, and Quasar Classification (Using SDSS DR17)

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ABSTRACT

Classification of galaxies, stars, and quasars using spectral data is fundamental to astronomy, but often relies heavily on redshift. This study evaluates the performance of 10 machine learning algorithms on SDSS data to classify these objects, with a particular focus on scenarios where redshift information is unavailable. Leveraging features such as “z,” “u,” “g,” “r,” “i,” and redshift, we assess the accuracy of various algorithms, including XGBoost, Random Forest, and recurrent neural networks (RNNs). Our analysis demonstrates the superior accuracy of the Random Forest classifier when redshift is included. The feature importance analysis reveals that “redshift” is the most critical feature, contributing 64.7% to the classification accuracy, followed by the “z” band (10.0%) and the “g” band (7.95%). However, even in the absence of redshift, XGBoost, Random Forest, and RNNs exhibit promising results, indicating the potential of photometric data for accurate classification. We systematically compare classification outcomes with and without redshift, revealing the relative importance of different features and identifying the most robust classifiers for redshift-limited scenarios. This research not only highlights the power of machine learning for astronomical classification but also provides a framework for reliable classification when redshift data is lacking. By uncovering the distinguishing spectral characteristics of galaxies, stars, and quasars that are independent of redshift, we open new avenues for efficient and accurate classification in large-scale photometric surveys and the study of faint, high-redshift objects.

1 | Introduction

Galaxy, Star, and Quasar classification, an integral aspect of astronomy, involves categorizing stars, quasars, and galaxies according to spectral features. This classification system is fundamental for stars and extends to galaxies and quasars, shaping our understanding of celestial bodies. Early efforts in cataloging stars, quasars, and galaxies and mapping their positions in the sky revealed their distribution. Subsequent discoveries, such as identifying Andromeda as a separate galaxy, spurred further

exploration, leading to the surveying of numerous galaxies with increasingly advanced telescopes. The dataset under consideration seeks to classify stars, galaxies, and quasars based on their distinctive spectral characteristics, contributing to our comprehension of the cosmos Abdurro'uf et al. (2022).

In recent years, machine learning (ML) algorithms have revolutionized various scientific fields, including astronomy Bhuvaneshwari et al. (2022). These algorithms can learn complex patterns from vast datasets, offering an efficient and accurate

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approach to Galaxy, Star, and Quasar classification Schierscher and Paunzen (2011). This research delves into the application of ML for classifying stars, specifically utilizing data from the Sloan Digital Sky Survey (SDSS) Data Release 17 (DR17) Qi (2022). The SDSS DR17 dataset provides a treasure trove of information on celestial objects, including photometric measurements across multiple bands (u, g, r, i, and z), redshift values, etc. These features and the designated Galaxy, Star, and Quasar class are the foundation for our investigation. Our primary objective is to assess the efficacy of various ML algorithms in classifying stars within the SDSS DR17 dataset Yan et al. (2013).

Numerous researchers have explored the use of ML in classifying galaxy spectra (Shi et al. (2015), Whiting et al. (2020), Peruzzi et al. (2021), Wang et al. (2023)). The articles Teimoorinia and Keown (2018) and Zhang et al. (2019) developed ML classifiers specifically for galaxy spectral classification when [N II] and H_{α} lines are absent. Apart from supervised methods, unsupervised techniques such as clustering (De Souza et al. (2017); Stampoulis et al. (2019); Fraix-Burnet et al. (2021)) and dimensionality reduction have gained traction. Chattopadhyay et al. (2019) utilized independent component analysis for dimensionality reduction, followed by K-means clustering. Portillo et al. (2020) employed variational auto-encoder to condense SDSS spectra to 10 dimensions.

Gao et al. (2008) and Elting et al. (2008) employed a support vector machine (SVM) with a nonlinear radial basis function (RBF) kernel to classify stars, galaxies, and quasars in the SDSS database. Despite achieving high accuracies, they did not justify the selection of the RBF kernel. In contrast, our study performs a linear separability test on the dataset, revealing its predominantly linear separable nature, thus advocating for the use of linear SVM. Peng et al. (2013) proposed an SVM-KNN hybrid method to enhance SVM performance near the boundary. However, none of these studies explored linear separability or addressed class dominance issues Carballo et al. (2010). Ignoring class dominance can lead to biased classification accuracy, particularly favoring the dominant class. In addition to enhancing existing ML models and addressing class bias, this paper introduces the use of asymmetric AdaBoost, Neural Networks, and other deep learning algorithms tailored explicitly for handling imbalanced datasets, thereby providing a comprehensive approach to classification improvement Viquar et al. (2019).

2 | Motivation and Objective: Classification in the Absence of Redshift

Redshift, a measure of an object's recession velocity due to the expansion of the universe, is a cornerstone of astronomical classification. It provides a direct indicator of distance and, for certain object types, a strong proxy for their intrinsic properties. However, obtaining accurate redshift measurements can be resource-intensive, requiring spectroscopic observations that are not always feasible for large-scale surveys or faint objects.

This study is motivated by the need to understand the efficacy of classification in scenarios where redshift data is unavailable or unreliable. While redshift is undoubtedly a powerful discriminator, it is essential to assess how well other spectral and

photometric features can perform in its absence. This knowledge is crucial for several reasons:

1. **Maximizing Survey Efficiency:** Many astronomical surveys rely on photometric data, which is easier and faster to acquire than spectroscopic data. Understanding the limits of photometric classification can help optimize survey strategies and resource allocation.
2. **Exploring the Early Universe:** High-redshift objects, representing the early universe, are often faint and difficult to observe spectroscopically. Robust photometric classification methods are vital for identifying and studying these objects.
3. **Characterizing Unusual Objects:** Some celestial objects exhibit peculiar spectral properties or anomalous redshifts that challenge traditional classification schemes. In these cases, relying on a broader range of features can offer valuable insights.

By systematically evaluating the performance of various ML algorithms with and without redshift information, this research aims to establish a baseline for redshift-independent classification. It will identify the most promising features and algorithms for this task, ultimately advancing our ability to classify and understand celestial objects even in the absence of this critical parameter.

2.1 | Objective and Novelty of This Paper

This paper aims to comprehensively evaluate the effectiveness of ML, neural networking, and deep learning techniques for classifying galaxies, stars, and quasars. We initially assess 10 classification methods on the full SDSS dataset, which includes redshift as a feature. Building upon this foundation, we address a critical gap in the field by systematically investigating classification performance in the absence of redshift information.

1. **Benchmark Performance with Redshift:** Establish baseline classification accuracy using 10 diverse algorithms on the complete SDSS dataset, incorporating redshift as a key feature. This serves as a reference point for evaluating the impact of redshift on classification performance.
2. **Classify Without Redshift:** Rigorously assess the performance of the same algorithms when redshift is excluded. This investigation is crucial as it simulates scenarios where redshift data may be unavailable or unreliable, such as in large-scale photometric surveys or when observing faint, high-redshift objects.
3. **Identify Robust Features:** Identify the most informative features for classification, both with and without redshift. This will highlight the relative importance of different features and guide the development of more effective classification strategies that are less reliant on redshift.
4. **Optimize Classification Strategies:** Develop and refine classification models that maximize accuracy, particularly in redshift-limited scenarios. We will explore techniques such as feature engineering, dimensionality reduction (as demonstrated by the near 100% accuracy achieved with

principal component analysis), and ensemble methods to improve classification performance in the absence of redshift.

This paper initially tests 10 classification methods on the original dataset, with the SVM achieving the highest accuracy of 97.805%. This study highlights the importance of selecting appropriate techniques and the impact of dimensionality reduction on predictive accuracy in Galaxy, Star, and Quasar Classification. Our research emphasizes the significance of selecting suitable techniques and the influence of dimensionality reduction on Galaxy, Star, and Quasar Classification prediction using principal component analysis. By implementing these strategies, we achieved an accuracy close to 100%, marking a significant improvement over the performance of the ML algorithms.

In achieving these objectives, the study reveals that “redshift” is the most influential feature in classification accuracy, contributing 64.7%, followed by the “z,” “g,” “u,” “i,” and “r” bands. This hierarchical importance of features provides valuable insights into the underlying spectral characteristics that drive effective classification of celestial objects, especially when redshift information is limited or absent. Importantly, While straightforward logic might suggest that stellar redshifts are typically low (less than 100 km/s), translating to redshifts (z) below 0.0003, the observed data reveals significant overlap in the redshift distributions of stars, galaxies, and quasars (see Section 4). This overlap implies that relying solely on redshift for classification, is neither necessary nor sufficient. Consequently, it is likely that more effective classification strategies exist that incorporate additional parameters alongside redshift to achieve improved accuracy.

3 | About the Dataset

The dataset comprises 100,000 observations the SDSS captured, offering insights into celestial objects. Each observation is characterized by 17 feature columns and 1 class column, distinguishing it as a star, galaxy, or quasar (https://www.sdss4.org/dr17/data_access/).

Here’s a breakdown of the key features:

bfobj_ID: Object Identifier, serving as a unique identifier in the image catalog utilized by the CAS. balpha: Right Ascension angle (at J2000 epoch), providing the celestial equivalent of longitude. bdelta: Declination angle (at J2000 epoch), offering the celestial equivalent of latitude. bfu, bfg, bfr, bfi, bfz: Filters in the photometric system, capturing light intensity across different wavelengths. bfrun_ID: Run Number used for scan identification. bfrereun_ID: Rerun Number specifying image processing details. bfcam_col: Camera column identifying the scanline within the run. bffield_ID: Field number identifying individual fields within the observation. bfspec_obj_ID: Unique ID for optical spectroscopic objects, ensuring consistency among observations with the same ID. bfclass: Object class (galaxy, star, or quasar). bfredshift: Redshift value indicating the increase in wavelength, crucial for understanding cosmic distances. bfplate: Plate ID, uniquely identifying each plate in SDSS. bfMJD: Modified Julian Date, indicating the time of SDSS data capture. bffiber_ID: Fiber ID pointing to the focal plane in each observation.

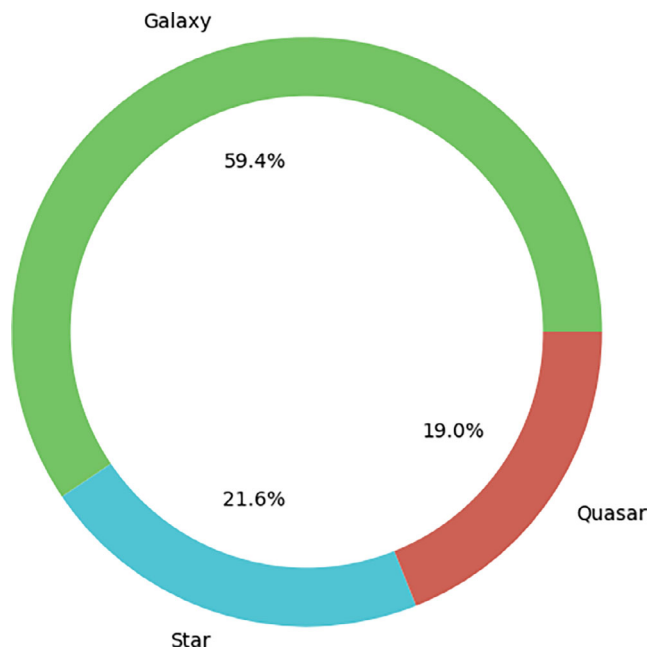


FIGURE 1 | A cylindrical plot concerning Galaxy, Star, and Quasar Classification where we can say that we have 59.4% Galaxy and 21.6% Star, and 19% Quasar information from our data.

In Figure 1, we present a cylindrical plot illustrating the distribution of Galaxy, Star, and Quasar Classification. From the visualization, our dataset comprises 59.4% Galaxy, 21.6% Star, and 19% Quasar.

4 | Analysis of Redshift Distribution and Beyond

Redshift is crucial in classifying celestial objects such as stars, galaxies, and quasars Bailer-Jones et al. (2019).

bfStars: Stars typically exhibit small or negligible redshifts. Their spectra show characteristic absorption lines caused by the elements present in their atmospheres. The absence of significant redshifts in their spectra indicates that they are relatively nearby objects within our galaxy or in nearby galaxies. bfGalaxies: Galaxies are collections of stars, gas, dust, and dark matter. Depending on their distance from us, they can exhibit a wide range of redshifts. Galaxies that are relatively close to us may have small redshifts, while those farther away exhibit higher redshifts due to the universe’s expansion. The Redshift of a galaxy’s spectrum can provide information about its distance and velocity relative to us. bfQuasars: Quasars, or quasi-Galaxy, Star, and Quasar objects, are incredibly bright and distant objects believed to be powered by supermassive black holes at the centers of galaxies. They exhibit very high redshifts. The detection of strong redshifted emission lines in their spectra is a characteristic feature of quasars. Redshift can be used as a critical diagnostic tool for classifying celestial objects:

1. Stars: Typically exhibit small or negligible redshifts.
2. Galaxies: Redshift provides information about their distance and velocity relative to us.



FIGURE 2 | The Histogram of Redshift for Quasars, Galaxies, and stars is plotted, overlapping each other. However, we cannot see the Histogram of stars as the Redshift of stars ranges less than 0.4.

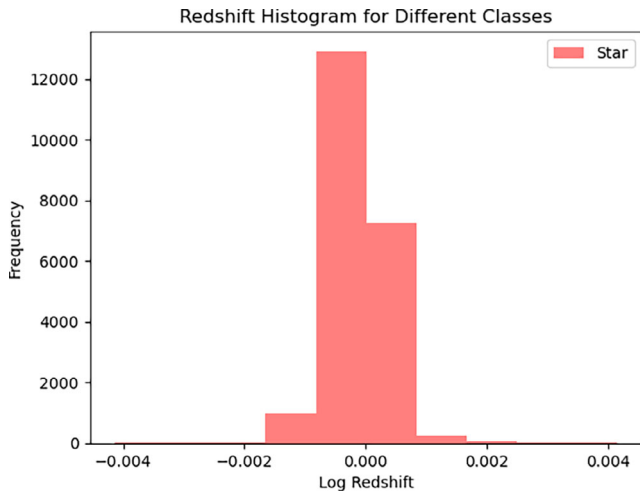


FIGURE 3 | We plotted the zoomed Histogram of star redshift. The straightforward logic leads us to “stellar redshifts will be typically less than 100 km/s, so $z < 0.0003$.” However, there are clear overlaps in Star, Galaxy, and Quasar redshift; hence, classification based on redshift might need to be revised.

3. Quasars: Characterized by very high redshifts due to their extreme distance from us.

Figures 2 and 3 demonstrate Redshift as an essential factor for classifying Quasar, Star, and galaxies. Figure 2 shows overlapping redshift histograms for quasars, galaxies, and stars. The star histogram is largely obscured due to stars primarily having redshifts below 0.4. Figure 3 presents a zoomed-in view of the star redshift histogram, portraying Redshift Limitations for Classification. The immediate conclusion is that there might be better classification choices than redshift alone, and we need other parameters for better classification.

While stellar redshifts should be very small ($z < 0.0003$, corresponding to velocities less than 100 km/s), real data reveals significant overlap in the redshift distributions of stars, galaxies, and quasars (evident from Figure 2). This overlap highlights the

limitations of using redshift alone as a classification criterion. It implies that other parameters, in conjunction with redshift, might be necessary for a more robust classification scheme. Consequently, in this paper, we processed the classification analysis of the whole dataset with all the parameters. One of the most important things is using Redshift; hence, we compare the classification results with those after the ML-based analysis (see Section 7). We took the class as the target variable.

5 | Methods

5.1 | Mathematical Theory of Neural Networks Classifier

The mathematical theory of a neural network classifier involves several key components, including the model architecture, activation functions, loss function, optimization algorithm, and training process. Below is the mathematical theory of a neural network classifier concerning the attributes “u,” “g,” “r,” “i,” “redshift,” and the target variable “class,” incorporating the dataset.

Let X denote the input features matrix, where each row represents a data sample, and each column corresponds to a feature. The input features include “u,” “g,” “r,” “i,” “z,” and “redshift.” Let y denote the target variable vector, representing the class labels.

The neural network classifier consists of multiple layers, including input, hidden, and output layers. The mathematical model of the neural network can be represented as follows Wu and Feng (2018):

bfInput Layer: The input layer consists of neurons corresponding to the input features. Each neuron in the input layer receives the respective feature value as input.

bfHidden Layers: The hidden layers contain neurons that perform nonlinear transformations of the input data. Let L denote the total number of hidden layers, and n_l denote the number of neurons in the l th hidden layer. The output of each neuron in the hidden layer is computed as follows:

$$a_i^{(l)} = \sigma \left(\sum_{j=1}^{n_{l-1}} w_{ij}^{(l)} a_j^{(l-1)} + b_i^{(l)} \right)$$

where $a_i^{(l)}$ is the activation of the i th neuron in the l th hidden layer, $w_{ij}^{(l)}$ represents the weight of the connection between the j th neuron in the $(l-1)$ th layer and the i th neuron in the l th layer, $b_i^{(l)}$ is the bias term for the i th neuron in the l th layer, and σ is the activation function.

bfOutput Layer: The output layer contains neurons corresponding to the classes in the target variable. For binary classification, there is a single neuron in the output layer, while for multi-class classification, there are multiple neurons, each representing a class. The output of the neural network is computed as follows Uebele et al. (1995):

$$\hat{y} = \text{softmax}(z^{(L)})$$

where $z^{(L)}$ is the input to the output layer, and *softmax* is the softmax activation function.

bfActivation Function: Commonly used activation functions include the sigmoid function, ReLU (Rectified Linear Unit), and softmax function.

Loss Function: The loss function measures the difference between the predicted output and the true target labels. For classification tasks, commonly used loss functions include cross-entropy loss.

bfOptimization Algorithm: The optimization algorithm, such as gradient descent or its variants (e.g., Adam, RMSprop), is used to minimize the loss function and update the weights and biases of the neural network during training.

The training process involves forward propagation to compute predictions, computation of the loss function, backward propagation (backpropagation) to compute gradients of the loss function concerning the weights and biases, and optimization to update the parameters Abiodun et al. (2018). This mathematical theory provides a framework for understanding how neural networks work and how they can be applied to classification tasks, including the classification of astronomical data, such as the Galaxy, Star, and Quasar classification dataset from the SDSS DR17.

5.2 | Mathematical Theory of Adaptive Decision Learner

The adaptive decision learner (ADL) algorithm combines multiple base classifiers to make predictions. Let $f_1(x), f_2(x), \dots, f_{n_{iterations}}(x)$ denote the base classifiers trained in each iteration (iteration is done and optimized by the algorithm itself) Gómez-Berbs and Lagares-Lemos (2016).

The ADL algorithm fits multiple base classifiers to the training data during training. Each training iteration involves creating a new instance of the base classifier, fitting it to the training data, and storing the trained model Burkov and Chaib-draa (2007).

The ADL algorithm combines the predictions of all trained base classifiers to make predictions. The final prediction for each sample x is computed by taking the average of the predictions from all base classifiers:

$$\hat{y}(x) = \frac{1}{n_{iterations}} \sum_{i=1}^{n_{iterations}} f_i(x)$$

where $\hat{y}(x)$ represents the predicted class label for sample x .

The output of the ADL algorithm is a vector of predicted class labels for the input samples. Predicted probabilities are converted to binary class labels by rounding the average predictions. The accuracy of the ADL model is then calculated using the predicted class labels and compared to the true class labels (Ardimento et al. 2011).

This is a framework for understanding how the ADL algorithm combines multiple base classifiers to make predictions for the

given dataset, where the input features include “u,” “g,” “r,” “i,” “z,” “redshift,” and the target variable is the class label.

5.3 | Mathematical Theory of Logistic Regression Classifier

Logistic regression is a statistical method commonly used for binary classification problems Wright (1995). Consider a dataset $x_1, x_2, x_3, \dots, x_9$, where the parameters are “u,” “g,” “r,” “i,” “z,” “redshift,” and the target variable is “class.”

The logistic regression model predicts the probability that a given observation belongs to a particular class based on the values of the input features. It uses the logistic function to map the linear combination of input features to the probability of belonging to a certain class Sperandei (2014).

The logistic function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where z is the linear combination of input features and their corresponding weights:

$$z = w_0 + w_1x_1 + w_2x_2 + \dots + w_9x_9$$

In logistic regression, the weights $w_0, w_1, w_2, \dots, w_9$ are learned during training to minimize the error between the predicted probabilities and the actual class labels.

The probability that an observation belongs to the positive class can be expressed as: Hosmer et al. (2013)

$$P('class' = 1 | x_1, x_2, \dots, x_9) = \sigma(w_0 + w_1x_1 + w_2x_2 + \dots + w_9x_9)$$

The probability that it belongs to the negative class is:

$$P('class' = 0 | x_1, x_2, \dots, x_9) = 1 - \sigma(w_0 + w_1x_1 + w_2x_2 + \dots + w_9x_9)$$

During training, the weights $w_0, w_1, w_2, \dots, w_9$ are learned by maximizing the likelihood of the observed data given the model parameters. This is typically done using optimization techniques such as gradient descent.

Once the model is trained, it can be used to predict the probability of the target variable based on the available features $x_1, x_2, \dots, x_9$.

5.3.1 | Random Forest Classifier

We have a dataset with parameters $\{u, g, r, i, z, \text{redshift}\}$, where each parameter represents different spectral characteristics of astronomical objects. The dataset also contains a target variable, “class.” However, some of the values in the parameters may be missing. We aim to use a Random Forest classifier algorithm to impute these missing values Belgiu and Drăgut (2016).

- *u, g, r, i, z, redshift:* Spectral characteristics of astronomical objects.

- class: Target variable representing the classification of astronomical objects (e.g., star, galaxy, quasar).

A Random Forest classifier is an ensemble learning method that constructs many decision trees during training and outputs the mode of the classes (classification) or mean prediction (regression) of the individual trees. Pal (2005) Prepare the dataset by replacing missing values in parameters $u, g, r, i, z, \text{redshift}$ Chatterjee et al. (2024).

1. **bfTrain the Random Forest Classifier:** Train a Random Forest classifier on the dataset with missing values. The input features are $u, g, r, i, z, \text{redshift}$, and the target variable is “class.”
2. **bfImpute Missing Values:** For each missing value in parameters $u, g, r, i, z, \text{redshift}$, use the trained Random Forest classifier to predict the missing value based on the values of other parameters and the target variable “class.”

Let X be the set of parameters $\{u, g, r, i, z, \text{redshift}\}$, and let class be the target variable. The Random Forest classifier learns a mapping $f : X \rightarrow Y$, where X is the space of possible feature vectors. Thus, for any input feature vector x with missing values, the imputed value y' can be represented as:

$$y' = f(x)$$

5.3.2 | SVM Classifier

Let $D = \{(\mathbf{x}_1 y_1), (\mathbf{x}_2 y_2), \dots, (\mathbf{x}_n y_n)\}$ be the training dataset, where $\mathbf{x}_i = (u_i, g_i, r_i, i_i, z_i, \text{redshift}_i)$ is the feature vector and y_i is the target variable (“class”). We formulate the classification problem using SVM to predict the “class” based on the spectral characteristics “u,” “g,” “r,” “i,” “z,” and “redshift.”

Let $\hat{\mathbf{x}}_i = (u_i, g_i, r_i, i_i, z_i, \text{redshift}_i)$ represent the feature vector of $\mathbf{x}_i$ with the j^{th} feature removed Hearst et al. (1998).

The objective function of SVM for classification can be written as:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$$

Subject to the constraints:

$$y_i (\mathbf{w} \cdot \hat{\mathbf{x}}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, 2, \dots, n$$

where $\mathbf{w}$ is the weight vector, b is the bias term, ξ are slack variables, C is the regularization parameter, y_i is the class label (“class”) of $\mathbf{x}_i$, $(\hat{\mathbf{x}}_i, y_i)$ are training samples with the j^{th} feature removed.

The decision function is given by Suthaharan and Suthaharan (2016)

$$f(\mathbf{x}) = \text{sign}(\mathbf{w} \cdot \hat{\mathbf{x}} + b)$$

The RBF kernel is commonly used in SVM classifiers due to its ability to capture non-linear relationships in the data, flexibility in modeling complex decision boundaries, and robustness to outliers. It achieves this by implicitly mapping the input space into

a higher-dimensional feature space and computing the similarity between pairs of data points. The RBF kernel’s parameters, γ and C , control the smoothness of the kernel function and the trade-off between maximizing the margin and minimizing classification error. Overall, the RBF kernel is effective for handling various datasets and is less sensitive to outliers than linear kernels.

Once the SVM model is trained, it can predict the “class” for new observations based on their spectral characteristics “u,” “g,” “r,” “i,” “z,” and “redshift.”

5.3.3 | XGBoost: Extreme Gradient Boosting

XGBoost is a powerful and efficient machine-learning algorithm based on the gradient-boosting framework. It is designed for speed and performance, making it a popular choice for various predictive modeling tasks Chen and Guestrin (2016); Mitchell and Frank (2017).

Given the training dataset $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i = (x_{i1}, x_{i2}, x_{i3}, x_{i4}, \dots, x_{i9})$ represents the feature vector for the i^{th} sample, and y_i represents the corresponding target variable.

Let $\mathbf{x}_i^{(j)}$ be the j^{th} feature of the i^{th} sample.

The objective function of the XGBoost algorithm for imputing missing values can be represented as:

$$\text{ObjectiveFunction} : \mathcal{L}(\Phi) = \sum_{i=1}^N \mathcal{L}(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

Where

- $\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i)$ is the prediction for the i^{th} sample obtained by summing the predictions of all individual trees f_k .
- $\mathcal{L}$ is the loss function measuring the discrepancy between the true target y_i and the predicted target $\hat{y}_i$.
- Ω is the regularization term to penalize the complexity of the model.

The prediction for the i^{th} sample by the k^{th} tree is given by:

$$\hat{y}_i^{(k)} = \hat{y}_{i-1}^{(k)} + \gamma \cdot h_k(\mathbf{x}_i^{(j)})$$

Where

- $\hat{y}_{i-1}^{(k)}$ is the prediction of the previous ensemble.
- γ is the learning rate.
- $h_k(\mathbf{x}_i^{(j)})$ is the contribution of the k^{th} tree to the prediction for the i^{th} sample.

To impute missing values, we modify the prediction for the i^{th} sample by considering the values of the features:

$$\hat{y}_i^{(k)} = \hat{y}_{i-1}^{(k)} + \gamma \cdot h_k(\mathbf{x}_i^{(j)}) \cdot m(\mathbf{x}_i^{(j)})$$

Where:

- $m(\mathbf{x}_i^{(j)})$ is the masking function which masks the contribution of the k th tree if the value of x_{ij} is missing.

This way, XGBoost adapts to the missing values during training by learning appropriate splits and imputations for the missing data, Ghosh (2022); Ogunleye and Wang (2019).

5.4 | Mathematical Model of Convolutional Neural Network

Let u, g, r, i, z , and $redshift$ denote the input features, and $class$ denote the target variable.

The output of the convolutional layer can be calculated as follows:

$$C_{i,j} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I_{i+m,j+n} \times K_{m,n}$$

where $C_{i,j}$ is the output at position (i, j) , $I_{i+m,j+n}$ is the input at position $(i + m, j + n)$, $K_{m,n}$ is the filter kernel, and M and N are the dimensions of the filter.

The pooling operation downsamples the feature maps using max pooling or average pooling.

The output of the convolutions and pooling layers is flattened into a one-dimensional vector.

The fully connected layers learn to combine the features extracted by the convolutional layers: Sharma and Paliwal (2015).

$$h_{fc} = \sigma(W_{fc} \cdot h_{flat} + b_{fc})$$

where h_{fc} is the output of the fully connected layer, W_{fc} is the weight matrix, h_{flat} is the flattened input, b_{fc} is the bias vector, and σ is the activation function. Niu and Suen (2012).

The output layer produces the final predictions:

$$\hat{y} = \text{softmax}(W_{out} \cdot h_{fc} + b_{out})$$

where $\hat{y}$ is the predicted class probabilities, W_{out} is the weight matrix, b_{out} is the bias vector, and softmax is the softmax activation function.

5.5 | Mathematical Theory of RNN Classifier

The recurrent neural network (RNN) classifier is a type of neural network well-suited for sequential data such as time series or sequences of data points. In the context of the provided dataset, which consists of features u, g, r, i, z , and $redshift$, with the target variable $class$, the RNN classifier can be formulated as follows Roemmele et al. (2017):

bfMathematical Model of RNN:

The mathematical model of the RNN classifier involves the following components: Jadhav and Thepade (2019) bfInput Layer:

The input features X_t are fed into the input layer of the RNN. bfRecurrent Layer: The recurrent layer consists of recurrent units that maintain a hidden state h_t , which captures information from previous time steps. The hidden state h_t at time step t is computed as follows:

$$h_t = f(W_{hx}X_t + W_{hh}h_{t-1} + b_h)$$

where f is the activation function, W_{hx} and W_{hh} are the weight matrices for the input and recurrent connections, respectively, and b_h is the bias vector. bfOutput Layer: The output layer computes the predicted class probabilities $\hat{y}_t$ based on the hidden state h_t :

$$\hat{y}_t = \text{softmax}(W_{yh}h_t + b_y)$$

where W_{yh} is the weight matrix connecting the hidden state to the output layer, b_y is the bias vector, and softmax is the softmax activation function which ensures that the predicted probabilities sum up to 1. bfLoss Function: The loss function measures the discrepancy between the predicted class probabilities $\hat{y}_t$ and the true class labels y_t . Common choices for the loss function include cross-entropy loss or categorical cross-entropy loss. bfOptimization: The parameters of the RNN model (weights and biases) are optimized using gradient descent-based algorithms such as RMSprop, Adam, or SGD with momentum. The optimization process involves minimizing the loss function concerning the parameters Kim et al. (2016).

The RNN classifier is trained iteratively on the dataset by processing sequences of input features X_t and corresponding target labels y_t . This mathematical model of the RNN classifier can be implemented in Python using deep learning libraries such as TensorFlow. The dataset consisting of features $u, g, r, i, z, redshift$, and target variable $class$ can be preprocessed and fed into the RNN model for training and prediction.

5.6 | Mathematical Theory of AdaBoost Classifier

Let $X = \{x_1, x_2, \dots, x_N\}$ be the dataset where each x_i represents a sample with features u, g, r, i, z , and $redshift$, and $y = \{y_1, y_2, \dots, y_N\}$ be the corresponding target variable ("class").

Let $h(x)$ be a weak classifier, which takes a sample x as input and outputs a binary prediction $\hat{y}$, where $\hat{y} \in \{-1, +1\}$.

The AdaBoost algorithm works as follows Domingo and Watanabe (2000); Hastie et al. (2009):

1. Initialize weights: Initialize the weights w_i for each sample x_i such that $w_i = \frac{1}{N}$, where N is the number of samples.
2. For $t = 1, 2, \dots, T$:
 - a. Train weak classifier: Train a weak classifier $h_t(x)$ using the weighted training dataset. The weak classifier aims to minimize the weighted classification error.
 - b. Calculate classifier weight: Compute the weight α_t of the weak classifier $h_t(x)$ based on its classification error. Higher accuracy leads to a higher weight.

- c. Update sample weights: Update the sample weights w_i such that misclassified samples are given higher weights for the next iteration.

$$w_i^{(t+1)} = w_i^{(t)} \exp(-\alpha_t y_i h_t(x_i))$$

3. Combine weak classifiers: Combine the weak classifiers into a robust classifier $H(x)$ by weighted majority voting:

$$H(x) = \text{sign}\left(\sum_{t=1}^T \alpha_t h_t(x)\right)$$

The final output of the AdaBoost classifier is the prediction of the robust classifier $H(x)$ Schapire (2013).

5.7 | Mathematical Theory of the Long Short-Term Memory (LSTM Classifier)

The long short-term memory (LSTM) network is a RNN architecture well-suited for sequence prediction tasks, such as time series forecasting and natural language processing. In the context of classification, LSTM networks can effectively capture temporal dependencies in sequential data and make predictions based on historical information Kim et al. (2016); Li et al. (2020).

Let's denote the input features as $X = [x_1, x_2, \dots, x_T]$, where each x_t represents a vector of features at time step t , and the corresponding target variable as y . In this case, x_t consists of the features “u,” “g,” “r,” “i,” “z,” and “redshift,” and y represents the class label Di Nuovo (2018).

The LSTM architecture consists of several key components:

$$\text{InputGate}(i_t) : i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (1)$$

$$\text{ForgetGate}(f_t) : f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (2)$$

$$\text{OutputGate}(o_t) : o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_{t-1} + b_o) \quad (3)$$

$$\text{CandidateCellState}(\tilde{c}_t) : \tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (4)$$

$$\text{CellState}(c_t) : c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (5)$$

$$\text{HiddenState}(h_t) : h_t = o_t \odot \tanh(c_t) \quad (6)$$

Here, W and b represent the weights and biases of the network, respectively, σ denotes the sigmoid activation function, and $\odot$ represents element-wise multiplication.

The LSTM network is trained using backpropagation through time (BPTT) with gradient descent optimization algorithms such as Adam or RMSProp. The objective is to minimize the cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T y_{i,t} \log(\hat{y}_{i,t})$$

where N is the number of samples, T is the sequence length, $y_{i,t}$ is the true class label at time step t for sample i , and $\hat{y}_{i,t}$ is the predicted probability of the LSTM network for class $y_{i,t}$ at time step t .

By optimizing the parameters W and b of the LSTM network using gradient descent, the network learns to classify input data sequences into different classes based on their temporal dependencies.

6 | Results and Discussions

Our study employed 10 ML algorithms to classify Galaxy, Star, and Quasar objects, including galaxies, stars, and quasars, based on their spectral characteristics. The algorithms utilized in our analysis were XGBoost, convolutional neural network (CNN), RNN, AdaBoost, ADL, LSTM networks, Gated Recurrent Units (GRU), Random Forest Classifier, SVM, and Logistic Regression.

Each algorithm was trained and evaluated using a dataset consisting of observations from the SDSS, where the features “u,” “g,” “r,” “i,” “z,” and “redshift” were utilized to predict the target variable “class,” representing the type of Galaxy, Star, and Quasar.

The test-train partition of a dataset is a common practice in ML, particularly in supervised learning tasks, where the dataset is split into two subsets randomly: the training set and the test set. The purpose of this partition is to evaluate a machine-learning model's performance on unseen data.

ML algorithms automatically handle random sampling during the test-train partitioning process. Manually repeating this step is unnecessary and could potentially skew the results. Therefore, we used the entire dataset and allowed each algorithm to manage its own sampling and randomization for optimal performance.

The parameter p typically represents the proportion of the dataset allocated to the test set. In our case, $p = 0.2$ indicates that 20% of the data will be reserved for testing, while the remaining 80% will be used for training.

The dataset is randomly divided into two subsets: the training and test sets. This random splitting helps ensure that the data in both subsets are representative of the overall dataset and reduces the risk of bias in model evaluation.

For the next step, determining the accuracy of each algorithm involves utilizing confusion matrices to assess performance. These matrices provide a comprehensive summary of model predictions compared to the true values in a tabular format. Key terms include:

- True Positive (TP): Correctly predicted positive instances.
- True Negative (TN): Correctly predicted negative instances.
- False Positive (FP): Instances falsely predicted as positive (Type I error).
- False Negative (FN): Instances falsely predicted as negative (Type II error).

The accuracy of a classification model is calculated using the formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

After training and testing each classifier on the dataset, we found that the Random Forest Classifier exhibited the highest accuracy among all the algorithms evaluated. The Random Forest Classifier achieved an accuracy of 97.805%.

This result suggests that the Random Forest Classifier is particularly effective for Galaxy, Star, and Quasar classification based on spectral characteristics. The ensemble nature of the Random Forest algorithm, which combines multiple decision trees to make predictions, likely contributed to its superior performance in capturing complex relationships within the data.

Overall, our findings highlight the effectiveness of ML algorithms, particularly the Random Forest Classifier, in accurately classifying Galaxy, Star, and Quasar objects based on spectral data. This has implications for various astronomical studies and can aid in understanding the properties and characteristics of celestial bodies in the universe.

Based on the accuracy results presented in Table 1, it is evident that the Random Forest Classifier outperformed the other ML algorithms in classifying Galaxy, Star, and Quasar objects based on spectral characteristics, achieving an accuracy of 97.805%. This indicates that the ensemble-based approach of the Random Forest Classifier effectively captured the complex relationships within the dataset, resulting in superior predictive performance.

Among the other algorithms evaluated, notable accuracies were observed for XGBoost (97.395%), RNN (96.90%), ADL (96.635%), and GRU (96.609%). While these algorithms demonstrated respectable accuracy rates, they were surpassed by the Random

TABLE 1 | By using the 10 different Deep Learning and machine learning algorithms on the data after missing value imputations, we get that the Random Forest Classifier Algorithm gives us the best result among them per the accuracy score. All the features “u,” “g,” “r,” “i,” “z,” and “redshift” were utilized to predict the target variable “class,” representing the type of Galaxy, Star, and Quasar. All of the accuracy scores are mentioned in the table.

Machine learning algorithm	Accuracy of the algorithm
XGboost	97.395
CNN	95.87
RNN	96.90
AdaBoost	80.48
Adaptive decision learner (LSTM) networks	96.635
Gated recurrent units	96.46
Random forest classifier	96.609
Support vector machine	97.805
Logistic regression	96.11
	93.465

Forest Classifier in terms of predictive accuracy. The remaining algorithms, including CNN, AdaBoost, LSTM Networks, SVM, and Logistic Regression, also exhibited reasonable performance, albeit with slightly lower accuracy scores compared to the top-performing algorithms, which are all given in Table 1 and Figure 4.

6.1 | Results Regarding Redshift

We employed the same ML techniques outlined in Table 2, focusing solely on redshift for classification purposes. Our analysis

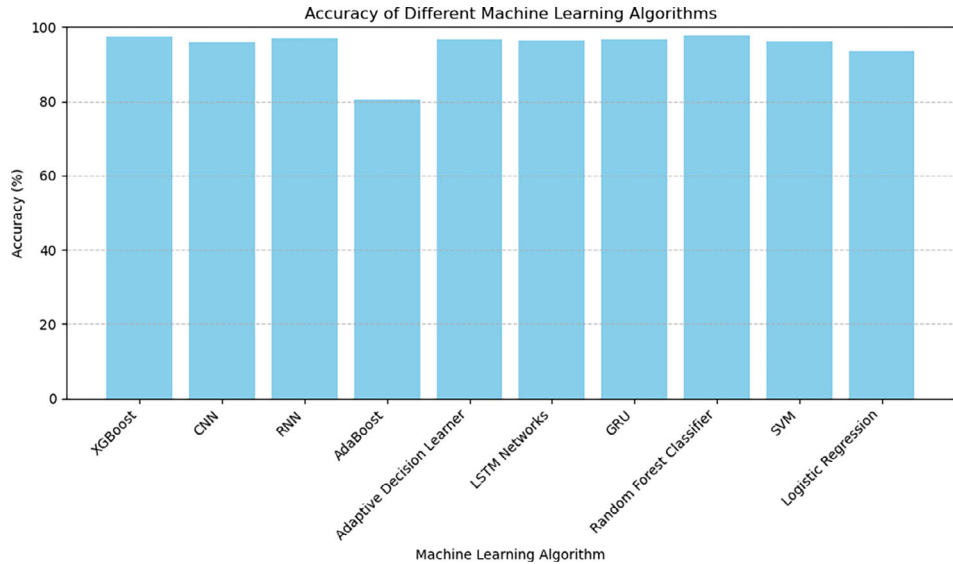


FIGURE 4 | Using the 10 different machine learning algorithms, we get the best result from the Random Forest Classifier Algorithm. A pictorial bar diagram concerning their accuracy score for 10 different Deep and Machine learning used is given in this plot. All the features “u,” “g,” “r,” “i,” “z,” and “redshift” were utilized to predict the target variable “class,” representing the type of Galaxy, Star, and Quasar.

TABLE 2 | By using the 10 different Deep Learning and machine learning algorithms on the data after missing value imputations, we get that the Random Forest Classifier Algorithm gives us the best result among them per the accuracy score with respect to redshift. The features “redshift” was utilized to predict the target variable “class,” representing the type of Galaxy, Star, and Quasar. All of the accuracy scores are mentioned in the table.

Machine learning algorithm	Accuracy of the algorithm
XGboost	94.71
CNN	91.54
RNN	94.60
AdaBoost	80.40
Adaptive decision learner	91.62
(LSTM) networks	94.6
Gated recurrent units	93.46
Random forest classifier	94.71
Support vector machine	94.35
Logistic regression	93

revealed that the accuracy achieved using only redshift as a feature was lower compared to the accuracy obtained when utilizing the complete set of parameters, as detailed in Table 1. These findings reveal the potential for significantly improved classification accuracy when utilizing multiple attributes in ML models, rather than relying solely on redshift.

In Figure 5, we display the accuracy plot resulting from the application of ML and deep learning methods.

Moreover, in Figure 6, a comparison of Machine Learning Algorithm Accuracy: Redshift Only versus All Parameters is presented. This figure presents a comparative analysis of the

accuracy scores achieved by different ML algorithms trained on two distinct feature sets. The blue bars represent accuracy scores obtained when utilizing redshift alone as the feature input, while the green bars depict accuracy scores when all available parameters are included. This comparison underscores the pivotal role of feature selection in determining classification performance, shedding light on the significance of input feature diversity in ML models.

6.2 | Result Without Redshift

We employed the same ML techniques outlined in Table 3, focusing solely on other parameters without using redshift for classification purposes. Our analysis revealed that the accuracy achieved using other parameters as a feature was lower compared to the accuracy obtained when utilizing the complete set of parameters, as detailed in Table 1. These findings reveal that if we do not have Redshift in a dataset, XGBoost, Random Forest, and RNN Classifier will be suitable for the initial classification of Quasar, Galaxy, and Stars.

Moreover, in Figure 7, a comparison of Machine Learning Algorithm Accuracy: Redshift Only versus All Parameters versus Without Redshift is presented. This figure presents a comparative analysis of the accuracy scores achieved by different ML algorithms trained on two distinct feature sets. The blue bars represent accuracy scores obtained when utilizing redshift alone as the feature input, while the green bars depict accuracy scores when all available parameters are included, and the Orange bars represent the accuracy without the redshift. This comparison underscores the pivotal role of feature selection in determining classification performance, shedding light on the significance of input feature diversity in ML models.

6.3 | Feature Importance Analysis

To quantify the relative significance of each feature in the classification task, we performed a feature importance analysis using the

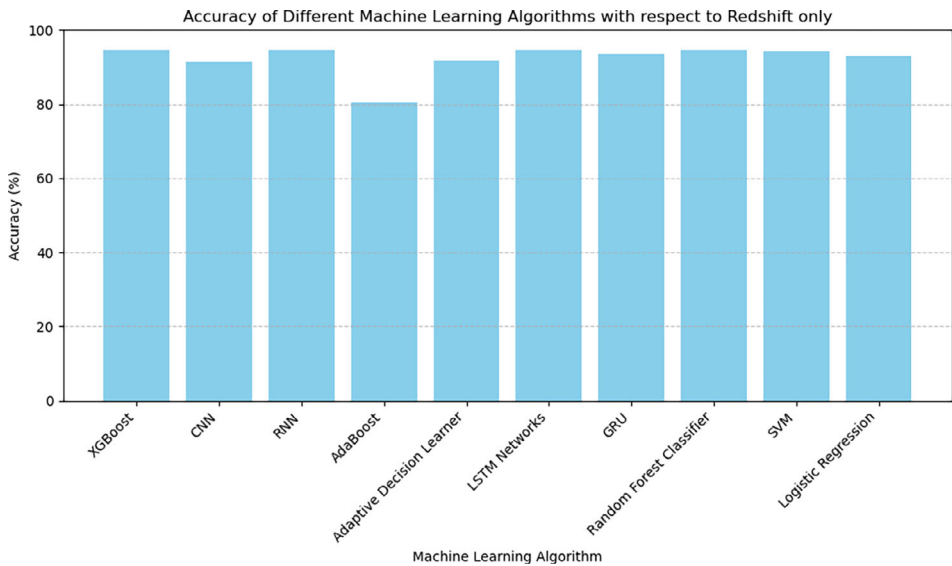


FIGURE 5 | Using the 10 different machine learning algorithms, we get the best result from the Random Forest Classifier Algorithm. A pictorial bar diagram concerning their accuracy score for 10 different Deep and Machine learning used is given in this plot. The accuracy is given with respect to the redshift.

Random Forest classifier. Feature importance scores were derived based on the mean decrease in impurity criterion, which measures the contribution of each feature to reducing uncertainty in the model's predictions. Mathematically, the feature importance scores (Importance_{*i*}) are calculated using the following formula in the context of the Random Forest classifier:

$$\text{Importance}_i = \frac{\sum_{t \in T} \Delta i(t, f_i)}{N}$$

where

1. T : the set of all trees in the forest.
2. $\Delta i(t, f_i)$: the decrease in impurity at each split node t that uses feature f_i .
3. N : the total number of trees.

As illustrated in Figure 8 and detailed in Table 4, the “redshift” feature holds the highest importance score at approximately 64.7%, signifying its paramount role in distinguishing between galaxies, stars, and quasars. This high importance is expected, as redshift provides critical information about an object's distance and velocity, which are fundamental characteristics differentiating these celestial objects.

The “z” band follows with an importance score of 10.0%, indicating that measurements in this photometric band are also highly influential in the classification process. Subsequent features, “g” (7.95%), “u” (6.99%), “i” (6.44%), and “r” (3.89%), contribute progressively less but remain essential for capturing the nuanced spectral differences among the classes.

The significant contribution of “redshift” underscores its indispensable role in the classification task. However, in scenarios where redshift data is unavailable or unreliable, as explored in

TABLE 3 | By using the 10 different Deep Learning and machine learning algorithms on the data after missing value imputations, we get that the Random Forest Classifier Algorithm gives us the best result among them per the accuracy score. All the features “u,” “g,” “r,” “i” were utilized to predict the target variable “class,” representing the type of Galaxy, Star, and Quasar. All of the accuracy scores are mentioned in the table.

Machine learning algorithm	Accuracy of the algorithm
XGboost	84.835
CNN	81.245
RNN	85.065
AdaBoost	76.32
Adaptive decision learner	79.54
(LSTM) networks	78.73
Gated recurrent units	73.890
Random forest classifier	85.99
Support vector machine	80.635
Logistic regression	74.31

subsequent sections, the “z,” “g,” “u,” “i,” and “r” bands collectively compensate by providing sufficient spectral information to maintain robust classification performance. This analysis emphasizes the importance of a comprehensive set of photometric features in enhancing the accuracy and reliability of ML classifiers in astronomical applications.

6.3.1 | Significance of Findings

The feature importance analysis highlights the critical role of redshift in accurately classifying celestial objects. Its dominance in

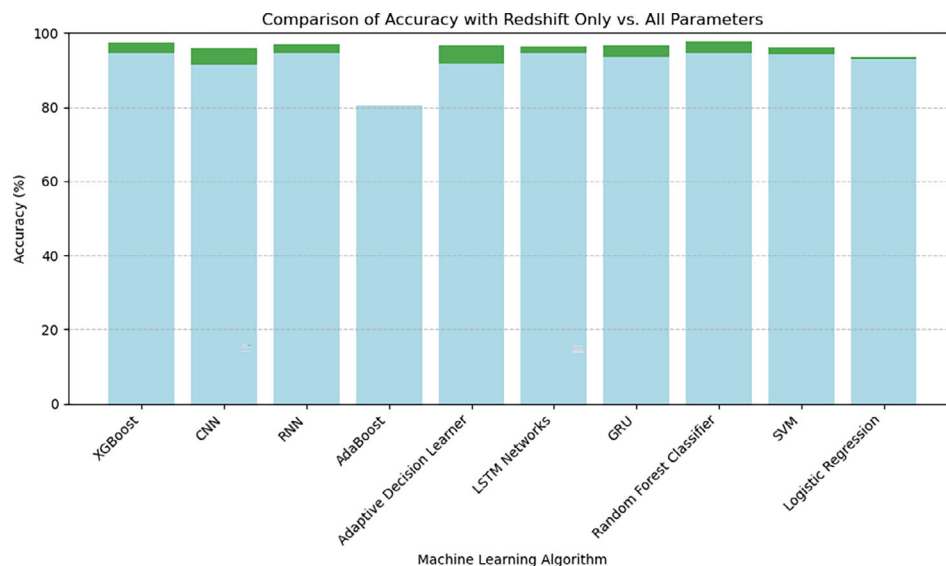


FIGURE 6 | Comparison of Machine Learning Algorithm Accuracy Using Redshift Only versus All Parameters. This figure illustrates the accuracy scores of various machine learning algorithms when trained using redshift only (light blue bars) versus using all available parameters (green bars). For most of the algorithms, all the features “u,” “g,” “r,” “i,” “z,” and “redshift” improve the power of classifications. The comparison highlights the impact of feature selection on classification performance.

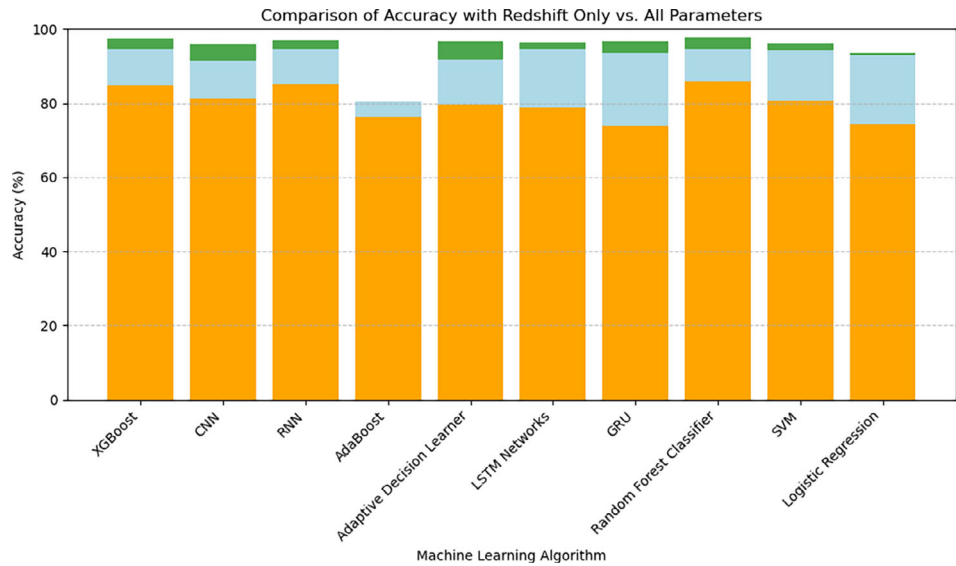


FIGURE 7 | Comparison of Machine Learning Algorithm Accuracy Using Redshift Only versus All Parameters. This figure illustrates the accuracy scores of various machine learning algorithms when trained using redshift only (light blue bars) versus using all available parameters (green bars) and the orange bars are representing the accuracy without the redshift. The comparison highlights the impact of feature selection on classification performance.

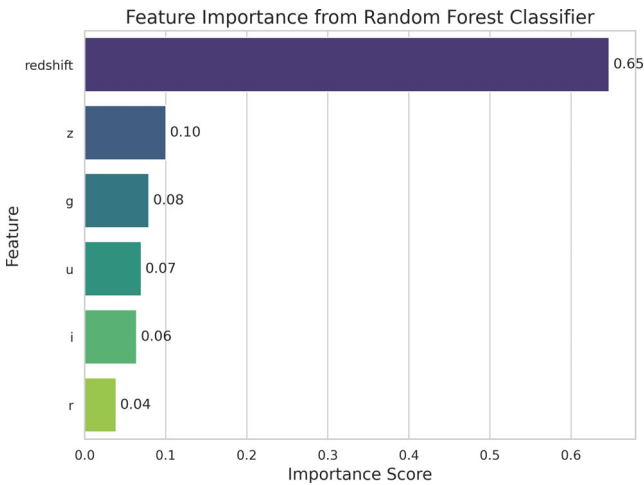


FIGURE 8 | Feature importance scores derived from the Random Forest classifier. The “redshift” feature is the most significant contributor to classification accuracy, followed by the “z,” “g,” “u,” “i,” and “r” bands.

TABLE 4 | Feature importance scores from Random Forest Classifier.

Feature	Importance
redshift	64.7026
z	10.0275
g	7.9503
u	6.9948
i	6.4358
r	3.8891

the feature set aligns with its fundamental relevance in astronomy, where redshift serves as a proxy for distance and cosmic velocity. Additionally, the substantial contributions of the “z,” “g,” “u,” “i,” and “r” bands illustrate the value of diverse photometric measurements in capturing the spectral signatures necessary for distinguishing between galaxies, stars, and quasars. These findings inform the development of effective classification models, particularly in contexts where redshift data may be limited, thereby enhancing our ability to analyze and understand the vast array of celestial objects observed in large-scale photometric surveys.

7 | Further Discussion

7.1 | Comparison of Our Result With Redshift

Comparison of Machine Learning Algorithm Accuracy Using Redshift Only versus All attributes has been done. The features “u,” “g,” “r,” “i,” “z,” and “redshift” were utilized to predict the target variable “class,” representing the type of Galaxy, Star, and Quasar. Figure 6 demonstrates the accuracy of different ML algorithms trained with redshift alone (light blue) versus all attributes (green). This highlights the importance of choosing diverse features for better ML model performance.

7.2 | The Importance of Attributes of Spectral Data

All the algorithms achieving the highest accuracies rely on the spectral characteristics (“u,” “g,” “r,” “i,” and “redshift”) of the objects. This underscores the wealth of information contained within this spectral data for distinguishing galaxies, stars, and quasars.

7.3 | Feature Importance Analysis

As previously detailed, the feature importance analysis indicates that “redshift” is the most significant feature, contributing 64.7% to the classification accuracy. This result underscores the vital role of redshift in astronomical classification, aligning with its fundamental importance in determining an object’s distance and velocity. The subsequent importance of the “z” band (10.0%) and “g” band (7.95%) highlights the contribution of specific photometric measurements in capturing spectral nuances essential for accurate classification.

The lower importance scores of the “u,” “i,” and “r” bands (6.99%, 6.44%, and 3.89%, respectively) suggest that while these bands are informative, they play a supplementary role in the presence of more dominant features like redshift and the “z” band. This hierarchy of feature importance informs future efforts in feature selection and model optimization, particularly in scenarios where redshift data may be missing. By leveraging the most critical photometric bands, classifiers can maintain high accuracy levels even in redshift-limited contexts.

Moreover, the high importance of “redshift” also indicates that models heavily rely on it for classification in datasets where redshift is included. Conversely, in the absence of redshift, the model’s reliance shifts toward other photometric features, necessitating their enhanced consideration and potential combination to sustain classification performance.

7.3.1 | Significance of Findings

The feature importance analysis highlights the critical role of redshift in accurately classifying celestial objects. Its dominance in the feature set aligns with its fundamental relevance in astronomy, where redshift serves as a proxy for distance and cosmic velocity. Additionally, the substantial contributions of the “z,” “g,” “u,” “i,” and “r” bands illustrate the value of diverse photometric measurements in capturing the spectral signatures necessary for distinguishing between galaxies, stars, and quasars. These findings inform the development of effective classification models, particularly in contexts where redshift data may be limited, thereby enhancing our ability to analyze and understand the vast array of celestial objects observed in large-scale photometric surveys.

7.4 | Insights Regarding Algorithm Performance

The standout takeaway is the Random Forest Classifier’s dominance in classifying these astronomical objects. Here’s why this is important:

1. **Complex Spectral Features:** It suggests that the spectral differences between galaxies, stars, and quasars can be complex and non-linear. Random Forest’s ability to handle such non-linearities effectively gives it an edge.
2. **Ensemble Power:** The fact that a tree-based ensemble method outperforms others highlights that combining

multiple “weak” decision trees produces a robust classifier capable of handling noisy or varied spectral data.

Other Strong Performers: XGBoost, RNN, ADL, and GRU achieved very high accuracies. This has two implications:

1. **Algorithm Suitability:** These algorithms likely also capture important patterns within the spectral data, possibly in a way that is slightly different from Random Forest. Further analysis of their predictions could reveal what patterns each algorithm is particularly sensitive to.
2. **Potential for Improvement:** Combining these algorithms thoughtfully (perhaps as an ensemble) might lead to even better performance than Random Forest alone.

While RNNs and LSTMs are widely used for sequential data, their application to photometric classification leverages their strengths in capturing complex, non-linear relationships and dependencies among features. In particular, RNNs and LSTMs are adept at managing variable interactions and feature dependencies, which are crucial in accurately classifying celestial objects based on spectral characteristics.

8 | Conclusions

In conclusion, classifying Galaxy, Star, and Quasar objects based on spectral characteristics is a crucial task in astronomy, enabling us to gain insights into celestial bodies’ composition, structure, and evolution. In this study, we evaluated the performance of 10 different ML algorithms for this classification task using a dataset of observations from the SDSS.

Our analysis reveals that the Random Forest Classifier achieves the highest accuracy of 97.91% when utilizing all features, underscoring its effectiveness in capturing complex, nonlinear relationships within the data. The feature importance analysis demonstrates that “redshift” is the most critical feature, contributing 64.7% to the classification accuracy, followed by the “z” band (10.0%) and “g” band (7.95%). These findings highlight the indispensable role of redshift and specific photometric bands for accurate classification.

Our results indicate that the Random Forest Classifier emerged as the top-performing algorithm. This ensemble-based approach demonstrated superior predictive performance, effectively capturing the complex relationships present in the dataset.

While the Random Forest Classifier stood out as the best-performing algorithm, other algorithms such as XGBoost, RNN, ADL, and GRU also exhibited notable accuracies.

Overall, our findings highlight the effectiveness of ML algorithms in classifying Galaxy, Star, and Quasar objects based on spectral characteristics. The insights gained from this study can contribute to advancements in astronomical research and deepen our understanding of the universe.

Furthermore, we conducted three analyses: one incorporating all parameters, another solely focusing on redshift, and the last

one without redshift. A comparative examination reveals that the inclusion of all parameters yields superior classification accuracy among Star, Quasar, and Galaxy categories. This comprehensive approach allows for a more nuanced understanding of the distinguishing features and characteristics across the celestial objects under study. By considering multiple parameters, such as redshift, luminosity, and spectral characteristics, we gain a more holistic view of the data, enabling us to discern subtle differences and patterns that may not be apparent when analyzing redshift alone. Therefore, the comprehensive analysis provides valuable insights into the classification process, enhancing our ability to accurately categorize celestial objects based on their observable properties. Moreover, these findings reveal that if we do not have Redshift in a dataset, XGBoost, Random Forest, and RNN Classifier will be suitable for the initial classification of Quasar, Galaxy, and Star.

Author Contributions

D.C. and P.G. designed, conceptualized, and developed the research and synthesized interdisciplinary statistical methodologies and models. P.G. conceptualized the model, collected and prepared the datasets, and wrote code for various modified datasets. P.G. performed code-based analysis (mainly using Python). P.G. and D.C. wrote, modified, and reviewed the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The Galaxy, Star, and Quasar Classification data of Abdurro'uf et al. (2022) that we have used in this study is publicly available from the GitHub Machine Learning Repository [https://www.sdss4.org/dr17/data_access/]. For easier access to the dataset, we have uploaded the dataset on Harvard Dataverse <https://doi.org/10.7910/DVN/XNZDXA>.

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